Bank questions on software Construaction

Lecture 1 – Software Construction

Question One: Multiple Choice Questions

1. Software construction mainly focuses on:
A. System deployment
B. Project management
C. Coding and debugging
D. Market analysis

Correct Answer: C

2. Which knowledge area is most strongly linked with Software Construction?
A. Software Economics
B. Software Design
C. Software Maintenance
D. Software Evolution

Correct Answer: B

3. Which of the following is NOT a goal of software construction?
A. Easy to understand
B. Ready for change
C. Safe from bugs
D. Maximizing code length

Correct Answer: D

4. Minimizing complexity mainly helps to:
A. Reduce readability
B. Improve understanding and maintenance
C. Increase code size
D. Delay testing

Correct Answer: B

5. Abstraction is used to:
A. Hide implementation details
B. Increase execution time
C. Remove documentation
D. Expose all data

Correct Answer: A

6. Constructing for verification focuses on:
A. Ignoring faults
B. Writing complex code
C. Making faults easy to detect
D. Avoiding testing

Correct Answer: C

7. Reuse in software construction helps to:
A. Increase development cost
B. Reduce software quality
C. Save time and reduce defects
D. Eliminate testing

Correct Answer: C

8. Which of the following is an example of a coding standard?
A. Hardware requirements
B. Variable naming conventions
C. Business rules
D. Project schedule

Correct Answer: B

9. Which life cycle model mixes design, coding, and testing?
A. Waterfall
B. Spiral
C. V-Model
D. Agile

Correct Answer: D

10. Incremental integration is preferred because it:
A. Delays testing
B. Makes error detection easier
C. Increases integration risk
D. Eliminates documentation

Correct Answer: B

11. Which of the following is NOT a key practical consideration during construction?
A. Coding style and readability

B. Error handling and resource management
C. User interface design and marketing strategy
D. Testing and integration

Correct Answer: C

12. Construction with reuse involves:
A. Building reusable libraries for the future
B. Using existing tested libraries in new software projects
C. Avoiding third-party assets
D. Rewriting similar modules manually

Correct Answer: B

13. Which of the following languages is used mainly for configuration rather than full application development?
A. C++
B. Java
C. YAML
D. LabVIEW

Correct Answer: C

Question Two: Complete the Following

1. The principle that avoids duplicating code is called __________.
Correct Answer: Code reuse

Question Three: Essay Questions

1. What principle avoids duplicating code?

2. Explain the main goals of software construction and how they affect software quality.

Lecture 2 – Static Analysis & Mutability

Question One: Multiple Choice Questions

1. Static analysis examines code:
A. During execution
B. After deployment
C. Without executing it
D. Only after compilation

Correct Answer: C

2. Java is classified as a:
A. Weakly typed language
B. Statically typed language
C. Dynamically typed language
D. Untyped language

Correct Answer: B

3. Static typing helps detect:
A. Runtime logic errors
B. Performance issues
C. Type mismatch errors
D. Hardware failures

Correct Answer: C

4. Which error is usually detected dynamically?
A. Wrong argument type
B. Syntax error
C. Divide by zero
D. Wrong method name

Correct Answer: C

5. One major benefit of static analysis is:
A. Higher runtime speed
B. Late bug detection
C. Early bug detection
D. Ignoring standards

Correct Answer: C

6. Which tool mainly enforces Java coding style rules?
A. JUnit
B. FindBugs
C. Checkstyle
D. JVM

Correct Answer: C

7. Which Java type is immutable?
A. ArrayList
B. String
C. StringBuilder
D. Array

Correct Answer: B

8. Which keyword prevents reassignment of a reference?
A. static
B. public
C. final
D. private

Correct Answer: C

9. Aliasing occurs when:
A. Objects are copied
B. Multiple references point to the same object
C. Variables are final
D. Objects are immutable

Correct Answer: B

10. Modifying a list while iterating over it causes:
A. Syntax error
B. Compilation error
C. Runtime exception
D. Logical warning

Correct Answer: C

11.Which of the following is an example of Static Checking?
A. Detecting divide-by-zero while the program runs
B. Detecting a syntax error before the program runs
C. Detecting memory leaks during execution

D. Logging runtime errors
Answer: B

12.Static typing means:
A. Types are checked at runtime
B. Types are checked at compile time
C. The programmer manually assigns memory
D. Variables can change type dynamically

Answer: B

Part B: Short Answer

11-What makes mutable objects risky?
Answer: Shared references

Part C: Essay Question

12-Explain static analysis and discuss how immutability helps reduce software defects.

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LECTURE 3: CODE REVIEW

1.What is the main purpose of code review?
A. To execute the program with test inputs
B. To evaluate source code quality without executing it
C. To rewrite existing code for performance
D. To compile the program

Answer: B

2.Code review improves which two aspects of software engineering?
A. Only the performance of the program
B. The programmer and the code itself
C. Only debugging speed
D. Only testing coverage

Answer: B

3.Which of the following is not a direct goal of code review?
A. Finding logical or style errors
B. Sharing knowledge among developers
C. Guaranteeing runtime speed improvements
D. Ensuring readability and maintainability

Answer: C

4.The DRY (Don't Repeat Yourself) principle means:
A. Avoiding comments in your code
B. Avoiding repeated code by using functions or methods
C. Keeping multiple versions of the same logic
D. Using identical variable names

Answer: B

5.In a good comment, you should explain:
A. What the code is doing
B. Why the code is written that way
C. How to fix compilation errors
D. The syntax of the programming language

Answer: B

"Documenting Assumptions" in functions means:
A. Writing what input and output the function expects and guarantees
B. Repeating code to make it explicit
C. Describing implementation details only
D. Using print statements for debugging

Answer: A

6."Fail Fast" principle means:
A. Letting errors accumulate until program crash
B. Detecting and handling problems immediately
C. Ignoring exceptions in early stages
D. Waiting until final testing to fix bugs

Answer: B

7.Which of the following is a "Magic Number"?
A. A variable like MAX_SPEED = 120
B. A hardcoded literal like if (speed > 120)
C. A descriptive constant name
D. A well-documented parameter

Answer: B

8. Which of the following naming styles is the most professional?
A. x1, temp, data
B. a, b, c
C. calculateAverage(), studentCount
D. doWork(), workDone()

Answer: C

9.Why should you avoid global variables? A. They slow down performance B. They reduce readability and increase coupling C. They can only store primitive types D. They prevent garbage collection

Answer: B

10.Code review helps make software:
A. Safe from bugs, easy to understand, and ready for change
B. Faster to execute but less maintainable
C. Shorter regardless of readability
D. Dependent on external tools

Answer: A

11.What principle prevents code duplication?
Answer: Don't Repeat

Essay Question 1

Explain the purpose of code review and discuss how it improves both code quality and developer skills.

12-What principle prevents code duplication?
Answer: Don't Repeat

LECTURE 4: TESTING

1.Choosing test cases based *only* on the function's specification, without looking at the actual implementation, is defined as:

A. Unit testing

B. Integration testing

C. Blackbox testing

D. Whitebox testing

Correct Answer: C

2.What is the term for a test that checks a combination of modules or the entire program?

A. Unit test

B. Path test

C. Regression test

D. Integration test

Correct Answer: D

3.In the context of testing, which type of coverage is generally considered *infeasible* to achieve 100% for a large program?

A. Statement coverage

B. Branch coverage

C. Path coverage

D. Assertion coverage

Correct Answer: C

4.The strategy of dividing the input space into sets of similar inputs where the program is expected to have similar behavior is known as:

A. Haphazard testing

B. Exhaustive testing

C. Partitioning

D. Whitebox testing

Correct Answer: C

5.Unit Testing focuses on:
A. Testing the whole system
B. Testing a single function or class in isolation
C. Testing database performance
D. Testing user interfaces only

Answer: B

6.Integration Testing focuses on:
A. Testing multiple modules working together
B. Testing only one method

C. Testing compiler performance
D. Testing documentation quality

Answer: A

7.Regression Testing is used to:
A. Detect new bugs introduced after changes
B. Verify GUI layout consistency
C. Reduce test execution time
D. Optimize runtime performance

Answer: A

8.What is the correct order in Test-First Debugging?
A. Fix bug → write test → rerun program
B. Write failing test → fix bug → keep test
C. Delete old tests → write new ones → refactor
D. Run program → delete bug → stop

Answer: B

9.Which statement best compares Code Review and Testing?
A. Code Review is automatic; Testing is manual
B. Code Review is human inspection; Testing is code execution
C. Both are fully automated D. Both require executing the program

Answer: B

10.Why are boundary cases important in testing?
A. They are usually the easiest to handle
B. Bugs often occur at the edges of input ranges
C. They make code run faster
D. They reduce the number of tests needed

Answer: B

11. Testing is part of which software engineering activity?
A. Design
B. Validation
C. Deployment
D. Documentation

Answer: B

Essay Question 1

Mention the defining characteristic of each of the following testing types: Blackbox and Whitebox testing.

Count: Stages of Test-First Debugging

Count and briefly describe the stages of the Test-First Debugging best practice.

Purpose of running a bug-reproducing test before fixing?

Lecture 5 - Debugging & Exceptions

1.Which of the following is an example of a technique used as the "first defense: make bugs impossible" at compile time?
A. Dynamic checking
B. Immutability
C. Static checking
D. Defensive programming

Correct Answer: C

2.In Java, what is the keyword used to declare a variable with an immutable reference?
A. static
B. private
C. const
D. final
Correct Answer: D

3.What is the core idea of defensive programming, of which checking preconditions is an example?
A. Refactoring code for better performance.
B. Anticipating potential errors and handling them proactively.
C. Using dynamic checking to eliminate runtime errors.
D. Ensuring every statement is covered by a test case.
Correct Answer: B

4.A serious problem with Java assertions is that they are turned off by default. For what scenario are they usually intended to be turned off?
A. During initial development and testing.
B. After the program is released to users.
C. When performing whitebox testing.
D. To avoid NullPointerException.
Correct Answer: B

5.Which statement is a key principle of encapsulation?
A. Dividing a system into separate components.
B. Using assertions with side-effects.
C. Keeping things private as much as possible, especially for variables.
D. Avoiding the use of the final keyword.
Correct Answer: C

6.What is an exception object handed off to the runtime system called?
A. The call stack
B. An exception handler
C. Throwing an exception
D. checked exception
Correct Answer: C

7. Which technique is considered the first defense against bugs according to the lecture?
A) Debugging using breakpoints
B) Exception handling
C) Static checking
D) Regression testing

8. Why is it considered good practice to declare method parameters as final?
A) To increase execution speed
B) To reduce memory usage
C) To prevent reassignment and document intent
D) To enable exception propagation

Answer: C

9. Which of the following best describes an assertion?
A) A mechanism for handling external failures
B) A replacement for exception handling
C) An executable statement that checks program assumptions
D) A statement that always executes in production code

Answer: C

10. Which design principle helps localize bugs by dividing a system into independent components?
A) Abstraction
B) Inheritance
C) Modularity
D) Polymorphism

Answer: C

11. Which of the following should NOT be checked using assertions?
A) Method preconditions
B) Method postconditions
C) Internal invariants
D) User input validity

Answer: D

12. When an assertion condition evaluates to false (and assertions are enabled), Java throws:
A) RuntimeException
B) IOException
C) AssertionError
D) IllegalArgumentException

Answer: C

1. An exception object is created and passed to the runtime system through a process called throwing an exception.
Answer: Throwing

2. The ordered list of method calls active at the time an exception occurs is called the call stack.
Answer: Call stack

3. The keyword used to explicitly generate an exception in Java is throw.
Answer: throw

Compare checked exceptions and unchecked exceptions in Java.
Include examples and explain when each should be used.

Discuss the role of modularity, encapsulation, and variable scope in localizing bugs.

Lecture 6 – Specifications

1.In the context of software contracts, the specification acts as a "firewall." What is the primary benefit of this firewall?
A. It forces the implementer to use Design by Contract (DbC).
B. It prevents all bugs from arising at the interface between components.

C. It results in decoupling, allowing the client and implementer to change their code independently as long as they adhere to the contract.
D. It ensures that every input is fully covered by the postcondition.

Correct Answer: C

2.The primary purpose of a software specification is to:
A) Describe how the code is implemented
B) Serve as a contract between the client and the implementer
C) Improve program performance
D) Replace testing

Answer: B

3.In the contract model of specifications, the precondition is an obligation on the:
A) Compiler
B) Runtime system
C) Implementer
D) Client (caller)

Answer: D

4.If a method is called when its precondition does not hold, the implementation:
A) Must still satisfy the postcondition
B) Must throw an exception
C) Is free to behave arbitrarily
D) Must return a default value

Answer: C

5.Behavioral equivalence between two methods depends primarily on:
A) Their source code
B) Their execution time
C) The client's expectations
D) The programming language

Answer: C

6.Which type of specification describes what the result should be, without describing how it is computed?
A) Operational
B) Declarative
C) Procedural
D) Algorithmic

Answer: B

7.A specification that allows multiple correct outputs for the same input is called:
A) Deterministic
B) Declarative
C) Under-determined
D) Operational

Answer: C

8.Why are declarative specifications generally preferred over operational ones?
A) They are faster to execute
B) They expose implementation details
C) They are shorter and implementation-independent
D) They are easier to test

Answer: C

9.In Java specifications, preconditions are typically documented using:
A) @return
B) @throws
C) @param
D) @override

Answer: C

10.Which of the following should a method specification NOT refer to?
A) Parameters
B) Return value

C) Side effects
D) Local variables

Answer: D

Compare deterministic and under-determined specifications with suitable examples.

Discuss the characteristics of a good specification.

1-A specification consists of two main parts: a ....and a ....

2-A specification that hides implementation details and focuses on results is called a ....

Lecture 7: AI in Software Development

1.The primary impact of Artificial Intelligence in software development is to:
A) Replace software engineers
B) Increase hardware efficiency
C) Enhance productivity, accuracy, and innovation
D) Eliminate software bugs completely

Answer: C

2.Large Language Models (LLMs) are primarily used in Generative AI to:
A) Optimize computer hardware
B) Create text and code generation applications
C) Replace databases
D) Manage operating systems

Answer: B

3.One major advantage of AI-driven bug detection is its ability to:
A) Remove the need for testing
B) Predict future bugs based on historical data
C) Guarantee bug-free software
D) Replace developers during debugging

Answer: B

4.Which of the following is NOT a typical application of AI in software development?
A) Code generation
B) Testing automation
C) Manual requirement interviews
D) Security enhancement

Answer: C

5.Prompt chaining is primarily used to:
A) Increase model size
B) Solve complex tasks by dividing them into smaller prompts
C) Reduce model accuracy
D) Replace unit testing

Answer: B

6.Which core AI technology primarily utilizes Natural Language Processing (NLP) to interpret human language descriptions and generate code suggestions?
A. Deep Learning
B. Machine Learning
C. Large Language Models (LLMs)
D. Generative AI (Gen AI)

Correct Answer: C

7.Generative AI (Gen AI) is defined as a subset of AI that focuses on which specific capability?
A. Simulating human learning and comprehension.
B. Analyzing historical data to predict future errors.

C. Systematically reviewing source code for non-compliance.
D. Creating original content such as text, images, video, audio, or software code in response to a user's prompt.

Correct Answer: D

8.An AI tool detects duplicate code or functions that are too long. This function falls under which specific AI application area?
A. Performance optimization
B. Code synthesis
C. Code review
D. Automated debugging

Correct Answer: C

9.In the Development & Testing phase of the SDLC, how does AI primarily accelerate the process for developers?
A. By replacing the need for unit testing entirely.
B. By speeding up coding, debugging, refactoring, and security checks through real-time suggestions.
C. By automatically handling all Level 1 support tickets.
D. By ensuring 100% statement coverage through exhaustive testing.

Correct Answer: B

10.The new software development paradigm where a developer uses natural language prompting to generate all necessary code is known as:
A. Agile Development
B. LLM-Native Development
C. Prompt Driven Development (PDD)
D. Deep Learning Development

Correct Answer: C

11.Which of the following security vulnerabilities, resulting from user input being directly concatenated into SQL strings, does AI specifically assist in detecting and fixing?
A. OutOfMemoryError

B. Cross-Site Scripting (XSS) and SQL Injection (SQLi)
C. NullPointerException
D. IndexOutOfBoundsException

Correct Answer: B

12.In the Maintenance and Support phase, AI contributes to long-term stability by continuously monitoring system metrics and logs to detect which of the following?
A. Code Synthesis errors.
B. Inefficient loops.
C. Anomalies like DDoS attacks or memory leaks.
D. Unused variables.

Correct Answer: C

1.Mention: LLM Success Variables

Mention the three variables that a developer must tune to achieve success when crafting a prompt for an LLM

2.Count and state the four key benefits of using Prompt Chaining (breaking a complex task into smaller, sequential prompts) over a single large prompt.

3.Explain how Generative AI impacts different phases of the Software Development Life Cycle (SDLC).

4.AI-powered tools that integrate directly into IDEs are known as ....

5.Breaking a complex task into sequential prompts is called..........

6. Generative AI uses .... to generate text and code.

Lec 8 :MACHINE LEARNING FOR SOFTWARE CONSTRUCTION

2. In traditional programming, the output is produced by providing the computer with:

A. Data only
B. Data and desired outputs
C. Program and output
D. Data and program

Correct Answer: D

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3. In machine learning systems, the role of the training process is to:
A. Explicitly write program rules
B. Optimize hardware resources
C. Learn the program from data
D. Replace datasets with models

Correct Answer: C

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4. In Tom Mitchell's definition of machine learning, the letter E refers to:
A. Error
B. Experience
C. Estimation
D. Environment

Correct Answer: B

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5. Which of the following best represents a performance measure (P) in text classification?
A. Number of documents
B. Size of vocabulary
C. Percentage of correctly classified texts
D. Type of classifier used

Correct Answer: C

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6. Training data that contains input–output pairs (x, y) is associated with:
A. Unsupervised learning
B. Reinforcement learning
C. Supervised learning
D. Clustering

Correct Answer: C

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7. Which of the following is an example of an unsupervised learning task?
A. Regression
B. Classification
C. Clustering
D. Policy learning

Correct Answer: C

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8. In reinforcement learning, the agent selects actions mainly to:
A. Reduce dimensionality
B. Discover data structure
C. Maximize cumulative reward
D. Minimize classification error

Correct Answer: C

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9. Feature extraction is performed during which step of building a machine learning system?
A. Validation
B. Representation
C. Estimation
D. Testing

Correct Answer: B

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10. In machine learning terminology, the true but unknown mapping from inputs to outputs is called:
A. Hypothesis function
B. Model class
C. Target function
D. Validation function

Correct Answer: C

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11. A good feature representation should be all of the following EXCEPT:
A. Robust
B. Discriminative
C. Highly correlated
D. Reliable

Correct Answer: C

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12. Using the test set multiple times for hyperparameter tuning is discouraged because it:
A. Increases model bias
B. Reduces training accuracy
C. Leads to overfitting on test data
D. Slows down convergence

Correct Answer: C

1.Explain the main types of Machine Learning with suitable examples.

2. Describe the main steps involved in building a Machine Learning system.

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1. Machine learning is a field of study that gives computers the ability to learn without being explicitly _________.
Correct Answer: programmed

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2. In supervised learning, the training data is provided as pairs of _________ and _________.
Correct Answer: (x, y)

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3. In unsupervised learning, the goal is to discover the _________ of the data.
Correct Answer: structure

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4. In reinforcement learning, the agent interacts with an __________ and receives __________ as feedback.
Correct Answer: environment, rewards

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5. Feature extraction is part of the __________ step in building a machine learning system.
Correct Answer: representation

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6. The unknown true mapping from input space X to output space Y is called the __________ function.
Correct Answer: target

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7. Hyperparameters are parameters that cannot be directly learned during the __________ process.
Correct Answer: training

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8. Using the test set multiple times for hyperparameter tuning is considered a very bad idea because it affects the model's __________ performance.
Correct Answer: generalization

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9. In K-Nearest Neighbor, the choice of the value of __________ is considered a hyperparameter.
Correct Answer: k

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1. **Cross-validation is mainly used to __________ hyperparameters when the training data size is small.**
   Correct Answer: tune

Lecture 9 – Introduction to Deep Learning

1. The main difference between traditional machine learning and deep learning is that deep learning:
A. Uses labeled data only
B. Requires manual feature extraction
C. Learns feature representations automatically
D. Works only for image data

Correct Answer: C

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2. Deep learning models are characterized by:
A. A single linear layer
B. Multiple layers of nonlinear processing units
C. Rule-based inference
D. Manual parameter tuning

Correct Answer: B

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3. The concept of *Hierarchical Compositionality* in deep learning refers to:
A. Combining datasets from multiple sources
B. Learning features at multiple levels of abstraction
C. Using shallow classifiers
D. Applying handcrafted features

Correct Answer: B

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4. The 2012 breakthrough that significantly revived interest in deep learning was:
A. ResNet
B. Transformer
C. AlexNet
D. GAN

Correct Answer: C

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5. The main role of activation functions in neural networks is to:
A. Reduce dataset size
B. Introduce non-linearity
C. Normalize input data
D. Prevent overfitting

Correct Answer: B

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6. Which activation function suffers most from the vanishing gradient problem?
A. ReLU
B. Leaky ReLU
C. Sigmoid
D. Softmax

Correct Answer: C

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7. The bias term in an artificial neuron mainly allows the activation function to:
A. Change learning rate
B. Shift left or right
C. Normalize inputs
D. Increase batch size

Correct Answer: B

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8. A single neuron with sigmoid activation is equivalent to:
A. Linear regression
B. Decision tree
C. Logistic regression
D. Support Vector Machine

Correct Answer: C

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9. Which neural network architecture is best suited for image recognition tasks?
A. RNN
B. CNN
C. LSTM
D. Transformer

Correct Answer: B

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10. The main purpose of pooling layers in CNNs is to:
A. Increase feature map size
B. Add non-linearity

C. Reduce dimensionality
D. Increase number of parameters

Correct Answer: C

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11. LSTM networks were introduced mainly to solve the problem of:
A. Overfitting
B. Vanishing gradients
C. High computational cost
D. Data imbalance

Correct Answer: B

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12. In Generative Adversarial Networks (GANs), the generator aims to:
A. Classify real data
B. Detect fake samples
C. Generate realistic fake data
D. Minimize classification loss

Correct Answer: C

Question Two: Essay Questions

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1. Define Deep Learning and explain how it differs from traditional machine learning.

2. Describe the structure of an artificial neuron and explain the importance of activation functions.

3. Explain the working principle of Convolutional Neural Networks (CNNs) and mention their main applications.

1. Deep learning models learn __________ representations of data through multiple layers.
Correct Answer: hierarchical

2. The activation function introduces __________ into a neural network.
Correct Answer: non-linearity

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3. The Sigmoid activation function outputs values in the range __________.
Correct Answer: 0 to 1

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4. ReLU activation function is defined as __________ for negative input values.
Correct Answer: zero

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5. CNNs use __________ layers to reduce the spatial dimensions of feature maps.
Correct Answer: pooling

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6. LSTM networks are a special type of __________ neural networks.
Correct Answer: recurrent

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7. Transformers rely mainly on the __________ mechanism instead of recurrence.
Correct Answer: self-attention

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8. GANs consist of two competing networks called the __________ and the __________.
Correct Answer: generator, discriminator